

Predictive Model Plan – Student Template

1. Model Logic (Generated with GenAI)

Model Type

Recommended Model: Logistic Regression

Model Logic

The proposed predictive model aims to identify customers who are likely to become delinquent by analyzing their financial and behavioral information. The dataset is first reviewed, and missing values are handled using appropriate imputation techniques. Categorical variables are converted into numerical form, while numerical features are scaled where necessary. The processed data is divided into training and testing sets before training a Logistic Regression model. After training, the model estimates the probability of delinquency for each customer and classifies them as either **high-risk** or **low-risk** based on a selected probability threshold. This allows Geldium to identify customers who may require early intervention.

Top 5 Input Features

- Credit Score
- Credit Utilization
- Missed Payments
- Debt-to-Income Ratio
- Income

2. Justification for Model Choice

Logistic Regression was selected because it is one of the most reliable and interpretable models for binary classification problems such as delinquency prediction. The model produces probability scores that help analysts understand why a customer is considered

high risk. It is computationally efficient, easy to implement, and suitable for financial institutions where transparency and regulatory compliance are essential. Compared to more complex models, Logistic Regression provides a strong balance between performance, explainability, and operational simplicity, making it an appropriate choice for Geldium's credit risk assessment process.

3. Evaluation Strategy

The effectiveness of the predictive model will be assessed using multiple evaluation metrics to ensure both accuracy and fairness.

Performance Metrics

- **Accuracy** – Measures the percentage of correct predictions made by the model.
- **Precision** – Indicates how many customers predicted as high risk actually become delinquent.
- **Recall** – Measures how effectively the model identifies customers who truly become delinquent.
- **F1 Score** – Provides a balanced measure of Precision and Recall.
- **AUC-ROC** – Evaluates how well the model distinguishes between delinquent and non-delinquent customers across different thresholds.

Fairness and Bias Evaluation

To ensure ethical decision-making, the model will be tested for potential bias across customer groups such as age, employment status, and location. Prediction results will be reviewed to confirm that no particular group is unfairly disadvantaged. If bias is detected, appropriate mitigation techniques such as re-sampling, feature review, or threshold adjustment will be applied.

Ethical Considerations

- Protect customer privacy and confidential information.
- Ensure transparency in model predictions.
- Prevent unfair discrimination during risk assessment.
- Regularly monitor model performance and retrain the model when new data becomes available.